Advanced Empirical Finance: Topics and Data Science

Multifactor models and portfolio sorts

Stefan Voigt

University of Copenhagen and Danish Finance Institute (DFI)

Spring 2027

Today

- Assignment 1 discussion (first hour)
- Sorting stocks on beta
- From one factor to many

Assignment 1 discussion

Sorting stocks on beta

Where the CAPM stands

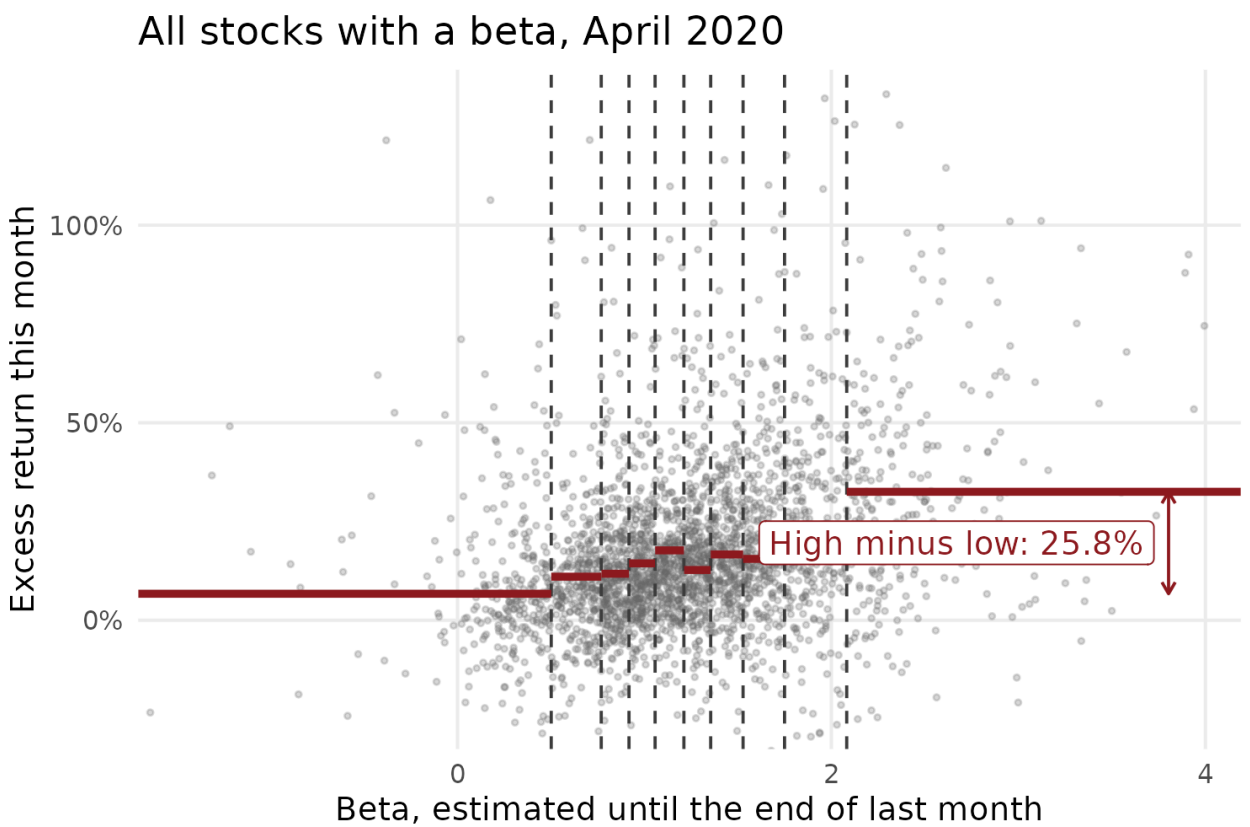
- **Lecture 6:** ten industries, 1960–2025. Higher betas came with barely higher average returns, but the GRS test could not reject the CAPM: the industries are too similar to each other and to the market
- **Today:** all US stocks in CRSP, sorted into 10 portfolios by their beta
- The betas are those of exercise 4: each month, a CAPM regression on the last 60 months (at least 48). That gives 2,401,783 betas for 18,498 stocks

> The CAPM says: the higher the beta, the higher the average return, one for one with the market premium

What a portfolio sort does

- Each month, rank the stocks by **last month's beta**, which investors know when they form the portfolios
- **Breakpoints** split them into 10 groups of equal size
- Each portfolio's return is the **value-weighted** average return of its stocks
- **High minus low:** the top portfolio minus the bottom portfolio

In April 2020 the market rose by 13.6%: high-beta stocks rose most. Repeat every month and average



Sorting on beta in code

R

```r
# Sort on last month's beta, which investors know when they form portfolios
beta_lag <- betas |>
  mutate(date = date %m+% months(1)) |>
  rename(beta_lag = beta)

beta_portfolios <- crsp_monthly |>
  inner_join(beta_lag, join_by(permno, date)) |>
  mutate(portfolio = ((row_number(beta_lag) - 1) * n_portfolios) %/% n() + 1, .by = date) |>
  summarize(
    ret_excess = weighted.mean(ret_excess, mktcap_lag),
    .by = c(portfolio, date)
  ) |>
  left_join(factors_ff3_monthly, join_by(date))
```

`betas` holds the betas of exercise 4 (`permno`, `date`, `beta`). Each month, the ranks of the betas, cut into ten equal parts, give the portfolio

Sorting on beta in code

Python

```python
# Sort on last month's beta, which investors know when they form portfolios
beta_lag = (
    betas.with_columns(pl.col("date").dt.offset_by("1mo"))
    .rename({"beta": "beta_lag"})
)

beta_portfolios = (
    crsp_monthly.join(beta_lag, on=["permno", "date"])
    .with_columns(
        portfolio=((pl.col("beta_lag").rank("ordinal") - 1) * n_portfolios // pl.len())
        .over("date") + 1
    )
    .group_by("portfolio", "date")
    .agg(ret_excess=(pl.col("ret_excess") * pl.col("mktcap_lag")).sum() / pl.col("mktcap_lag").sum())
    .join(factors_ff3_monthly, on="date", how="left")
)
```

`betas` holds the betas of exercise 4 (`permno`, `date`, `beta`). Each month, the ranks of the betas, cut into ten equal parts, give the portfolio

The CAPM for each portfolio

R

```
capm <- beta_portfolios |>
  nest(data = -portfolio) |>
  mutate(
    fit = map(data, \(d) summary(lm(ret_excess ~ mkt_excess, data = d))$coefficients),
    alpha = map_dbl(fit, \(f) f["(Intercept)", "Estimate"]),
    se_alpha = map_dbl(fit, \(f) f["(Intercept)", "Std. Error"]),
    beta = map_dbl(fit, \(f) f["mkt_excess", "Estimate"]),
    mean = map_dbl(data, \(d) mean(d$ret_excess))
  ) |>
  select(portfolio, alpha, se_alpha, beta, mean) |>
  arrange(portfolio)
```

The same regression as for the industries in lecture 6, now for the 10 beta portfolios

The CAPM for each portfolio

Python

```
rows = []
for (portfolio,), data in beta_portfolios.group_by("portfolio"):
    fit = smf.ols("ret_excess ~ mkt_excess", data=data.to_pandas()).fit()
    rows.append({"portfolio": portfolio, "alpha": fit.params["Intercept"],
                 "se_alpha": fit.bse["Intercept"], "beta": fit.params["mkt_excess"],
                 "mean": data["ret_excess"].mean()})
capm = pl.DataFrame(rows).sort("portfolio")
```

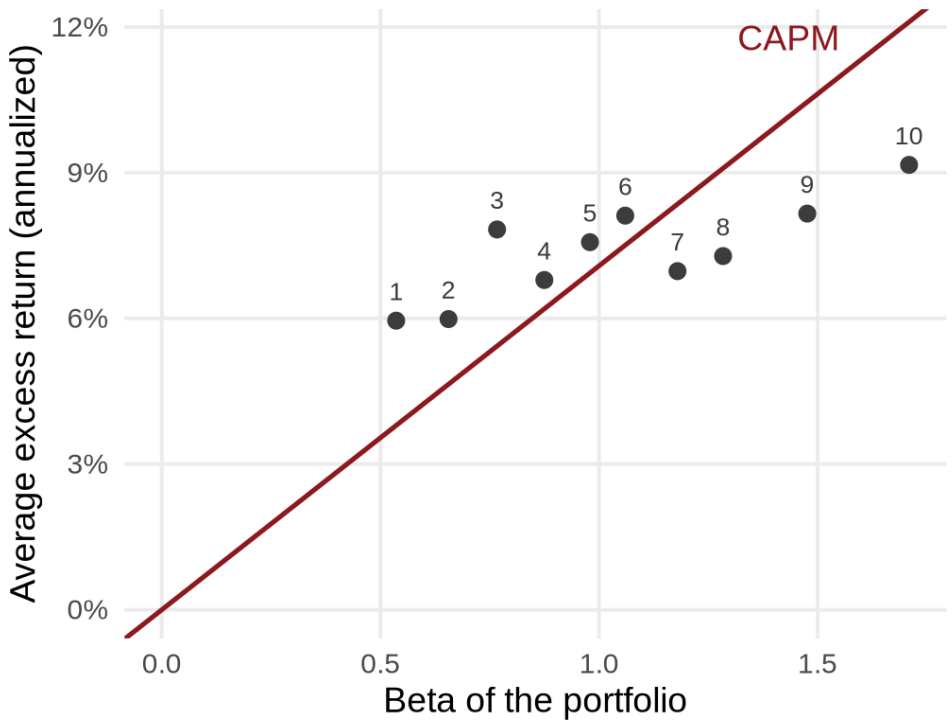
The same regression as for the industries in lecture 6, now for the 10 beta portfolios

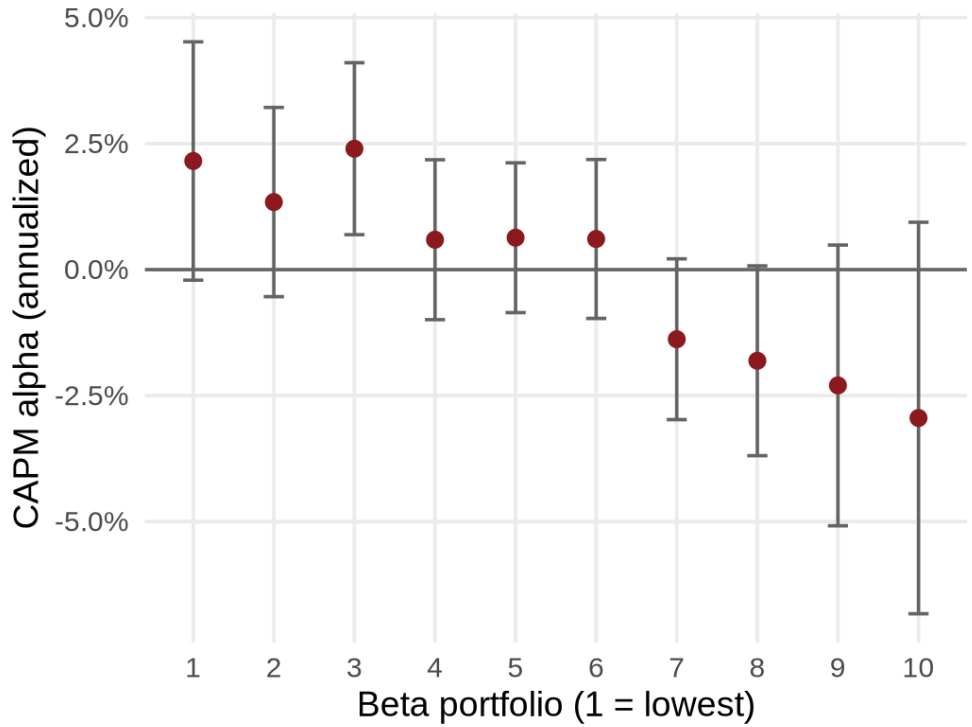
Which stocks have the higher alpha?

Ten portfolios sorted on beta, 1964–2025: which have the higher CAPM alpha?

- High-beta stocks
- Neither
- Low-beta stocks

Ten beta portfolios





- Betas from 0.54 to 1.71, average excess returns only from 6.0% to 9.2% a year. The CAPM predicts 3.8% to 12.1%: the line is too flat
- High minus low: alpha -5.1% a year (t = -1.86). GRS = 1.28, p-value 0.24: a clear pattern, but 1964–2025 is not enough to reject the CAPM with ten correlated portfolios

From one factor to many

Whiteboard: many factors

Suppose the SDF is linear in K factors, $m = a - b'f$. What does $E[m\,R_i^e] = 0$ imply for expected returns?

$$E[R_i^e] = \beta_i'\lambda, \qquad \beta_i = \mathrm{Var}(f)^{-1}\mathrm{Cov}(f, R_i^e), \qquad \lambda = \frac{\mathrm{Var}(f)\,b}{E[m]}$$

- If the factors are excess returns, each factor prices itself: $\lambda = E[f]$. In $R_{i,t}^e = \alpha_i + \beta_i' f_t + \varepsilon_{i,t}$, the model says $\alpha_i = 0$ for every asset, and GRS tests all alphas at once
- **Arbitrage pricing theory** (Ross, 1976): if returns have a factor structure and there is no arbitrage, expected returns are approximately $\beta_i'\lambda$

Where do factors come from?

- We do not observe betas. A characteristic z_i (size, book-to-market, past returns) is useful when it tells us something about the betas: $E[R_i^e] = \beta(z_i)'\lambda$
- Value firms are often close to distress and lose most in bad states: high beta, high premium
- A **portfolio sort turns a characteristic into a factor**: the top portfolio minus the bottom portfolio. Fama & French (1993) add SMB (small minus big) and HML (high minus low book-to-market) to the market: the three-factor model
- Today's sort on beta itself: low-beta stocks earn too much, high-beta stocks too little. Frazzini & Pedersen (2014) find the same in many markets and asset classes and turn it into a factor, *betting against beta*

Anomalies

Anomaly	Finding	Key reference(s)
Betting against beta	α declining in β across many asset classes	Frazzini & Pedersen (2014)
Size	Small stocks outperform large stocks	Banz (1981); Fama & French (1992)
Value	High book-to-market stocks outperform growth stocks	Rosenberg, Reid & Lanstein (1985); Fama & French (1992)
Momentum	Intermediate-term winners outperform losers	Jegadeesh & Titman (1993); Carhart (1997)
Liquidity	Illiquid stocks earn a return premium	Pástor & Stambaugh (2003)

- Hundreds more have been published since: the *factor zoo*. Which of them survive is a question for the lecture on reproducibility; how to use many characteristics at once, for the machine learning lectures